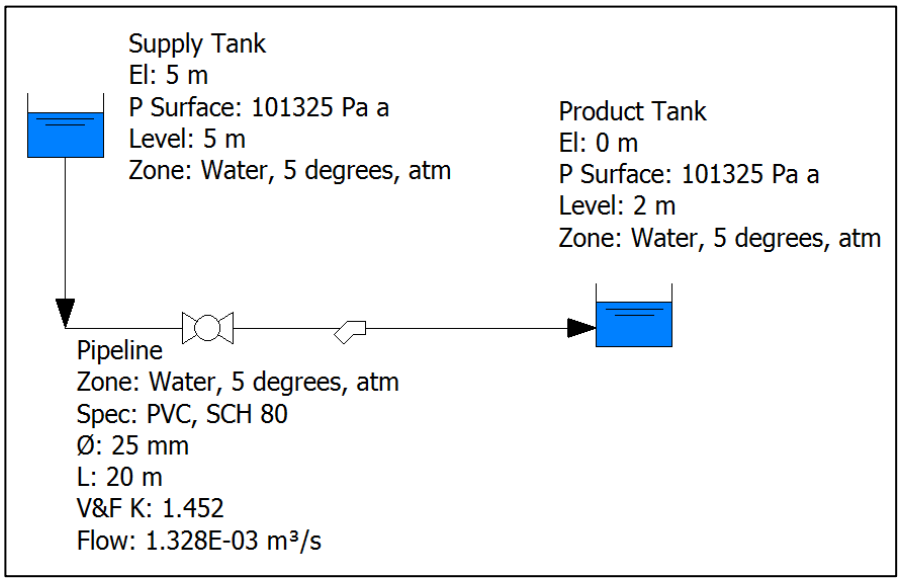
ChE 2O04 Tutorial #5 – Introduction to PIPE-FLO

Ensure that PIPE-FLO Advantage is installed on your computer, fully activated and all the units are SI.

NOTE: This tutorial is NOT a substitute for studying the full PIPE-FLO courseware that is posted on Avenue-to-Learn; it is simply meant to give you a quick introduction to the software and its intricacies.

→ We will be introducing how to make a simple system on PIPE-FLO including making fluid zones, specifications, and using devices such as tanks, valves, bends, etc.
→ The FLO-SHEET for the system as shown below is posted on Avenue-to-Learn.



V&F K: 90° Mitre Bend and fully opened Ball Valve

Questions:

1) Using your intuition, which way would the water flow, what forces are acting on the fluid?

2) What is the flow rate of water in the pipeline connecting the two tanks? What is the average velocity of the water in the pipeline?

3) What happens if you change the liquid level in the Product Tank to 20 m? Why?

Tutorial Activities

1) Setup the system at the default conditions shown in the image above and then explore/report the effect of the various process changes described below on the flow rate in the pipe connecting to the two tanks.

a) How does the flowrate change with pipe length (20, 40, 60 m) and pipe diameter (25, 32, 40 mm)? Report the volumetric flowrate and Reynolds number for each and comment and whether the flow is laminar or turbulent. If none of the cases show laminar flow, propose a set of values that would, based on the observed trends.

Volumetric flowrate (m^3/sec) at a supply tank liquid level of 5 m:

	Pipe Diameter (mm)		
Length of pipe (m)	**25**	**32**	**40**
20			
40			
60			

Reynolds number (-) at a supply tank liquid level of 5 m

	Pipe Diameter (mm)		
Length of pipe (m)	**25**	**32**	**40**
20			
40			
60			

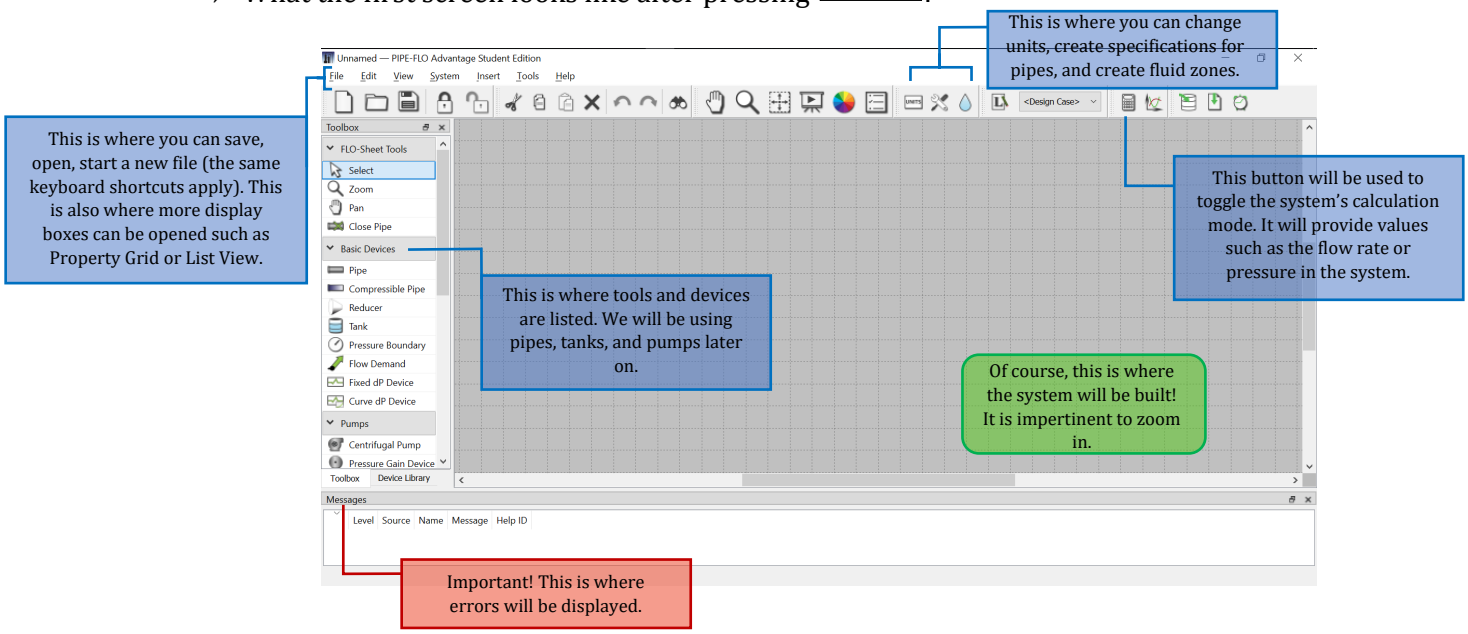
b) How does the above trends change with the supply tank liquid level? Using a pipe length of 20 m and a pipe diameter of 25 mm, how does the Reynolds Number and Flowrate change at a supply tank liquid level of 5, 10, and 15 m?

	Values at a pipe length of 20 m and a diameter of 25 mm	
Supply Tank Liquid Level (m)	**Volumetric flowrate (m^3/s)**	**Reynolds number (-)**
5		
10		
15		

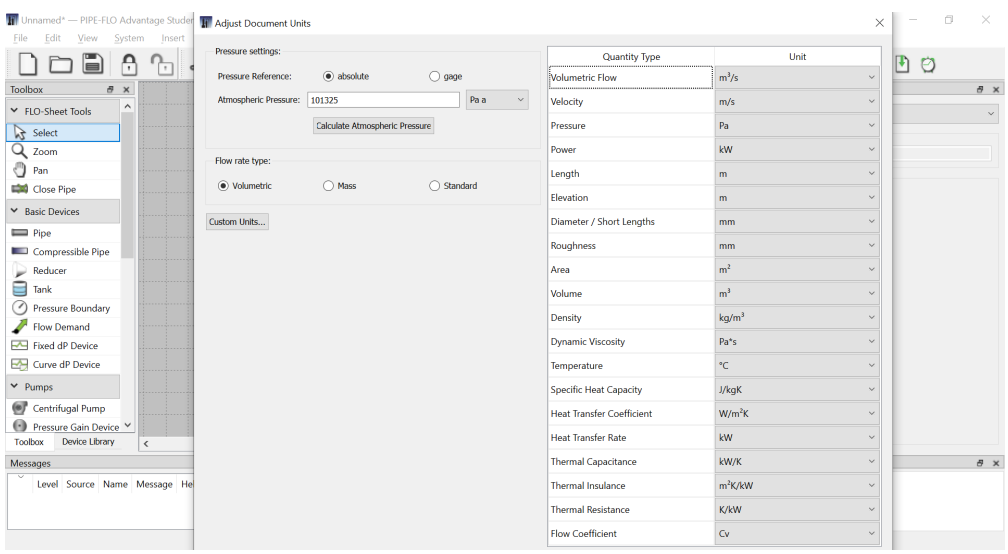
c) Based on the trends that you've identified in the tutorial, what is the combination of values in the above three variables that provides the highest flow rate? (**Hint: you can use any combination of the pipe diameter, length, and supply tank liquid level values we used today**) Present some reasons for why that is.

What we should address in the introduction:

→ What the first screen looks like after pressing New:



→ The first thing that needs to be done is changing the pressure from gauge to absolute and changing all the units to SI.



→ Next, it is important to cover the types of errors that can come about when building a system. PIPE-FLO uses levels to share the urgency:

Level 0 Pre-calculation check or calculation engine errors are preventing the program from calculating any results (*e.g.* physically-impossible parameters entered, like a negative absolute pressure). If a Level 0 message is generated, no other level messages will be displayed because no results have been calculated.

Level 1 Even though results may be calculated, the results are invalid or a system configuration is invalid (*e.g.* reversed flow or missing parameters). The results are unequivocally wrong and require close evaluation.

Level 2 The calculated results exceed either the limits of the equipment or limits of the equations used to obtain the results (*e.g.* the calculated flow rate through a pump exceeds the range for which data is available). There is increased uncertainty with regards to the accuracy of the displayed results.

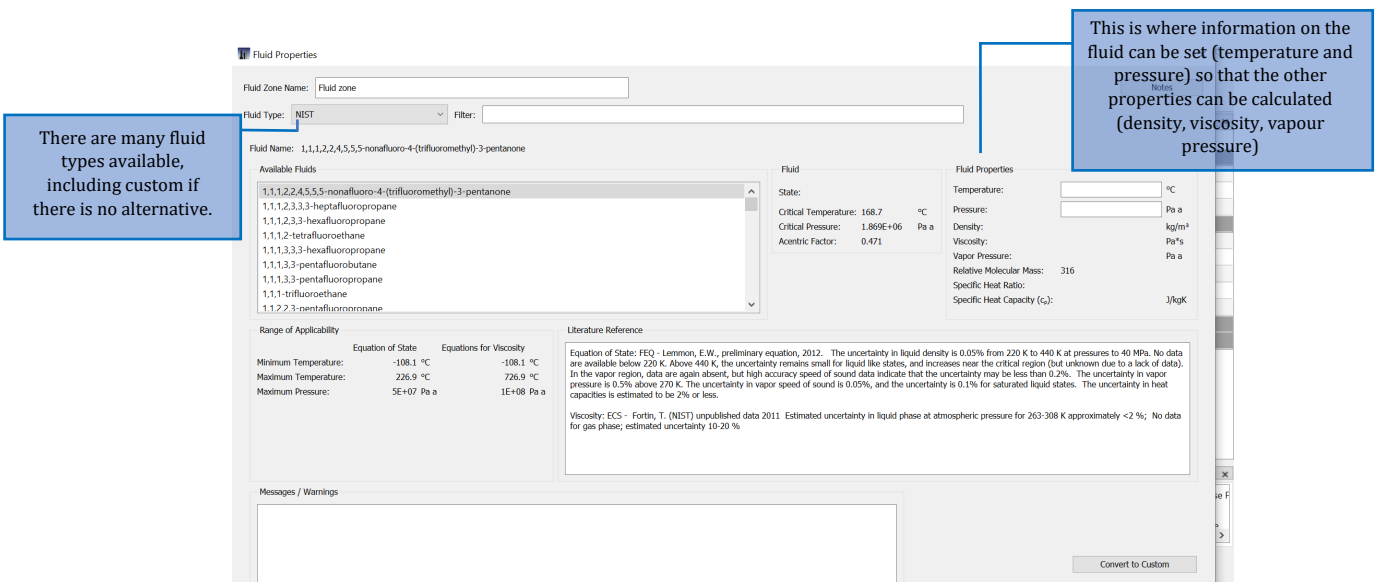
Level 3 The calculated results exceed specified operational limits (*e.g.* the pressure in a tank exceeds its design limits).

Level 4 This level constitutes informational messages that may or may not be based on the calculated results. These messages provide information about the status of devices (*e.g.* the number of iterations, the validity of laminar flow assumptions).

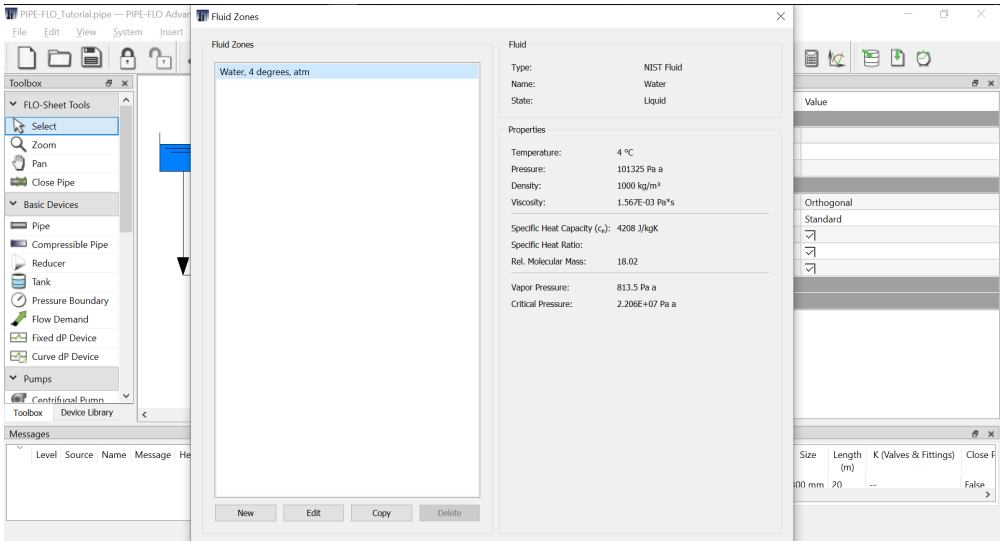
→ Save the file as the default project named "PIPE-FLO_Tutorial", this way all the settings will stay the same for any subsequent projects.

→ To begin creating the system, first we will create the fluid zone. Click the droplet in the top bar to begin, click new:

- Fluid Zone Name: Water, 4 degrees, atm
- Fluid Type: NIST
- Available Fluids: Water
- Fluid Properties, Temperature: 4 degrees
- Fluid Properties, Pressure: 101325 Pa



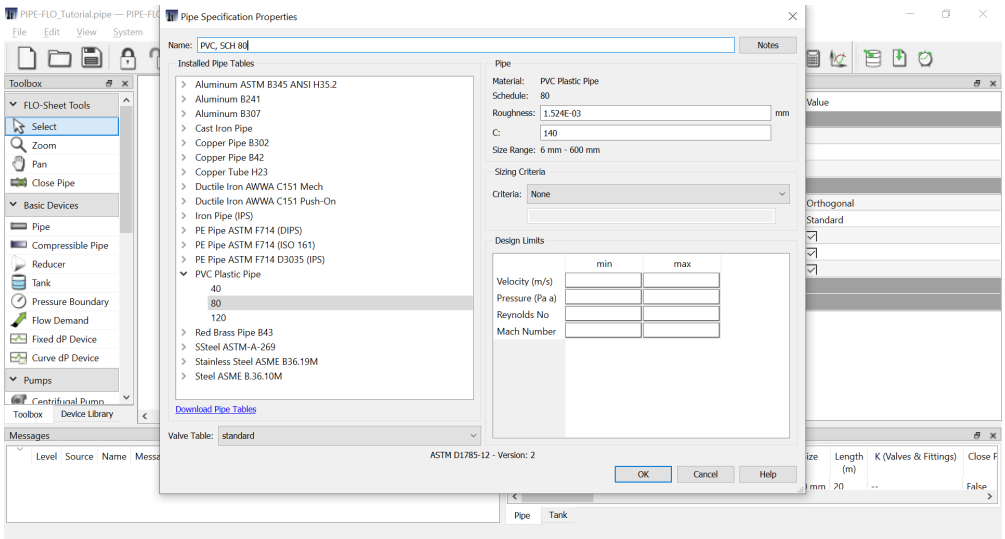
→ Once finished, it should look like this:



→ The next step is to create the pipe specification. Press okay and click the wrench and screwdriver in the top bar to begin, click new:

- Name: PVC, SCH 80
- Installed Pipe Tables: ∨ PVC Plastic Pipe, 80

→ Once finished, it should look like this:



→ Next, we will start placing basic devices.

→ Underneath the toolbox panel on the left, select tank and place it on the grid. Place another tank onto the grid and then press Esc to alight the device from the cursor. *It does not actually matter where the tank is placed, only the properties will dictate how the fluid behaves. However, keeping the system to scale is a visual method of making sure everything is working correctly.*

→ Click on Tank 1 to access and change its properties.
- Name: Supply Tank
- Elevation: 5 m
- Surface Pressure: 101325 Pa a
- Liquid Level: 5 m
- Fluid Zone: Water, 4 degrees, atm
- Everything else can be left unspecified.

→ Click on Tank 2 to access and change its properties.
- Name: Product Tank
- Elevation: 0 m
- Surface Pressure: 101325 Pa a
- Liquid Level: 2 m
- Fluid Zone: Water, 4 degrees, atm
- Everything else can be left unspecified.

→ Underneath the toolbox panel on the left, select pipe and hover over the Supply Tank. You will notice arrows, click on the tank to attach the pipe to it.

→ Keep dragging the pipe until the other end is attached to the Product Tank. You can click the grid if you need to bend the pipe.

→ Click on Pipe 1 to access and change its properties.
- Name: Pipeline
- Fluid Zone: Water, 4 degrees, atm
- Specification: PVC, SCH 80
- Size: 100 mm
- Length: 20 m
- K (Valves & Fittings): > Bend, Mitre Bend @ 90°, add, then > Valves and add Ball, then the value 1.038 should be displayed.
- Everything else can be left unspecified.

→ Now that the system has been built, it can be verified with the image above and parameters can be calculated using F5 or pressing the calculator button . Please answer the attached questions.